COMP-4150

Conceptual & Logical Design

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Team Members

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Relational Schema

1. **AUTHORS** (AuthorID, AuthorName)
2. **CATEGORIES** (CategoryID, CategoryName)
3. **USERS** (UserID, Name, Email, PasswordHash, Role, AccountStatus)
4. **BOOKS** (BookID, ISBN, Title, Description, Price, Quantity, AuthorID*, CategoryID*)
5. **BOOK_COPIES** (CopyID, Barcode, CopyStatus, BookID*)
6. **ORDERS** (OrderID, OrderDate, DeliveryType, UserID*)
7. **ORDER_ITEMS** (OrderID*, BookID*, Quantity, PriceAtPurchase)
8. **LOANS** (LoanID, BorrowDate, DueDate, ReturnedDate, OverdueFee, LoanStatus, UserID*, CopyID*)

Note: Underlined Attributes are Primary Keys, and Asterisk (*) represent attributes that are Foreign Keys

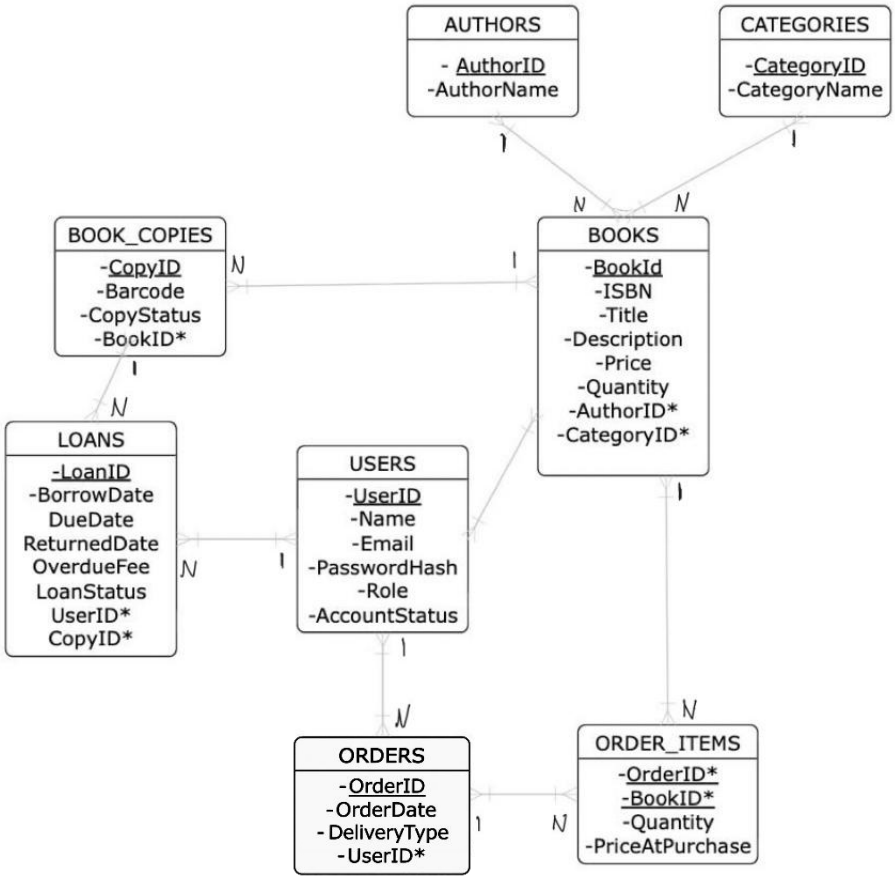
Additional Constraints

USERS: Email must be UNIQUE to prevent duplicates.

BOOKS: ISBN must be UNIQUE, this is to identify different books

BOOK_COPIES: Barcodes must be UNIQUE to identify each copy of a book

ER Diagram



Note: The software used to make this diagram has the lines show the type of relation (1 à N, 1 à 1, etc.). I thought the lines were too light to properly see those indications, so I opted to handwrite 1 and N to show the relations instead of relying on that.

Please ignore the crow's feet on the lines, those do not accurately represent the relations between entities.

Normalization Steps

1NF: All 9 of the tables above satisfy 1NF because each attribute stores a single value, each table has its' own primary key, and there's no repeating groups.

2NF: The tables satisfy 2NF because they are in 1NF and their non-key attributes depend on the entire primary key. i.e. in ORDER_ITEMS, Quantity and PriceAtPurchase depend on both OrderID and BookID.

3NF: All tables satisfy 3NF because they are in 2NF and non-key attributes depend directly on the primary key rather than on other non-key attributes